

Underwater Vision-Based Surveillance Using ML and AI

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Day-14: Peer Visual Review

1. Objective

To review the dataset analysis, annotation quality, preprocessing, split strategy and enhancement work completed during Week 2, and to define the hypotheses and experimental plan for the next stage.

2. Dataset and Annotation Findings

Three datasets were analysed separately:

- DIATAquarium: 11 classes, 8,121 training images and 773 validation images. The local test set is unavailable.
- Aquatic Plant: 1 class, 1,892 training images, 179 validation images and 89 test images.
- Well: 4 classes, 3,795 training images, 191 validation images and 184 test images.

Important findings:

- YOLO annotation validation found 0 invalid annotation entries across all three datasets.
- Aquatic Plant has 13 training images without matching labels.
- Well has 15 training images without matching labels.
- Some of these images appeared to be valid background/no-object images, so they were not automatically removed.
- Aquatic Plant had 6 duplicate images in the checked training sample.

3. Image Quality and EDA Findings

The visual EDA identified common underwater image problems:

- Haze and low visibility
- Low contrast
- Colour variation/distortion
- Dark regions or borders

The sampled image resolutions were:

- DIATAquarium: 1920×1080 in training and 2048×1536 in validation
- Aquatic Plant: 640×640
- Well: 640×640

Basic statistics from 100 sampled training images per dataset:

Dataset	Avg. Brightness	Avg. Contrast
DIATAquarium	104.51	49.10
Aquatic Plant	141.71	31.98
Well	125.52	38.01

These results show that the three datasets have different image-quality characteristics.

4. Class Distribution Findings

Class imbalance was particularly significant in the Well dataset:

Class	Object Count
fishes	50,194
school-of-fish	653
stone	5
Inlet-pipe	3

- DIATAquarium also contains uneven class frequencies, while Aquatic Plant contains only one class.
- Rare classes may therefore be more difficult for the detector to learn.

5. Split and Source-Level Findings

Source-level analysis identified:

- DIATAquarium → 6 sources
- Aquatic Plant → 2 sources
- Well → 1 source
- Images from the same source occur across different splits, creating a potential source-level leakage risk because frames from the same source can be highly similar.
- A completely source-independent split is difficult for Aquatic Plant and Well because they have only 2 and 1 source respectively.
- The existing supplied splits were preserved for Dataset V1 and the limitation was documented.
- No video data is currently available, so tracking is not included in the current implementation.

6. Dataset V1 and Preprocessing

Dataset V1 was prepared while keeping the original raw datasets unchanged.

Final preprocessing:

Raw Image → Resize to 640×640 → RGB Conversion → Normalize Pixel Values to 0–1 → Model Input

Training augmentation was defined as:

- Horizontal flip
- Small rotation
- Scale variation
- Brightness/contrast variation

Augmentation is applied only during training.

7. Enhancement Baseline

The classical enhancement baseline selected was: White Balance + CLAHE

- White Balance → reduces colour distortion.
- CLAHE → improves local contrast.

The enhanced images were stored separately so that the original images remain available for baseline comparison.

A total of 13,808 training images were enhanced:

- DIATAquarium → 8,121
- Aquatic Plant → 1,892
- Well → 3,795

The corresponding labels were also prepared and verified.

8. Original vs Enhanced Experiment

The main experiment will compare:

Original Images → YOLO → Baseline Results

against:

Original Images → White Balance + CLAHE → Enhanced Images → YOLO → Enhanced Results

The comparison will be performed separately for all three datasets.

Evaluation metrics:

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95
- Inference time

9. Experiment Hypotheses

H1: White Balance + CLAHE will improve colour balance and local contrast in degraded underwater images.

H2: Enhanced images may improve object-detection performance, especially under low-visibility and low-contrast conditions.

H3: Enhancement may not improve every dataset or class, because excessive colour or contrast changes can also affect useful features.

H4: Rare classes in the Well dataset are expected to be more difficult to detect because of severe class imbalance.

H5: The effect of enhancement is expected to differ across the three datasets because their image characteristics and class distributions are different.

10. Week-2 Final Outcome

During Week 2, the three datasets were audited, annotation and image-quality issues were identified, visual EDA was completed, Dataset V1 was prepared, preprocessing and augmentation were defined, source-level leakage risks were documented, and the White Balance + CLAHE enhancement baseline was implemented.

The original-vs-enhanced YOLO comparison protocol and experiment hypotheses are now defined, setting up the Week-3 model development and comparative experiments.